

Move & Learn: Motor Skills Development App

Persona:

Akbar is an 8-year-old boy with mild cerebral palsy that affects the coordination of his arms. While he can walk and move independently, he finds tasks requiring precise or fast hand movements challenging. Akbar enjoys watching cartoons and playing simple tablet games but often loses interest in repetitive therapy exercises. He needs engaging, interactive activities that feel like play while supporting his motor skills development. Bright visuals, positive sound feedback, and adaptive gameplay that matches his pace help keep him motivated and confident. At home, Akbar's parents encourage his therapy, and his physical therapist, Ms. Sara, closely monitors his progress. Ms. Sara values tools that allow her to assign customized tasks remotely and easily track Akbar's performance over time. Together, they benefit from an easy-to-use, adaptive app that transforms motor skill therapy into a fun, motivating experience for Akbar and a practical, insightful resource for his support team.

Name: Akbar, Age 8

- **Condition:** Mild cerebral palsy affecting arm coordination
- **Tech Exposure:** Loves cartoons and simple tablet games
- **Behavior:** Gets bored with repetitive therapy
- **Needs:**
 - Enjoyable games that feel like play, not therapy
 - Clear visual cues and sound feedback
 - Adaptive difficulty that matches his abilities
- **Frustrations:**
 - Games that require fast or complex gestures
 - Lack of motivation to do daily exercises

Stakeholder Persona:

Name: Ms. Sara, Pediatric Physical Therapist

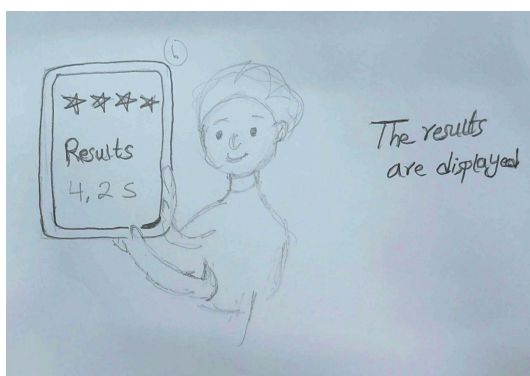
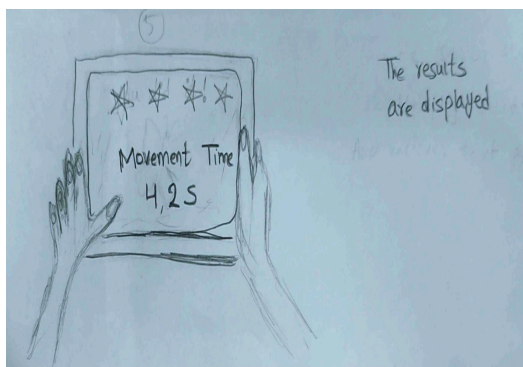
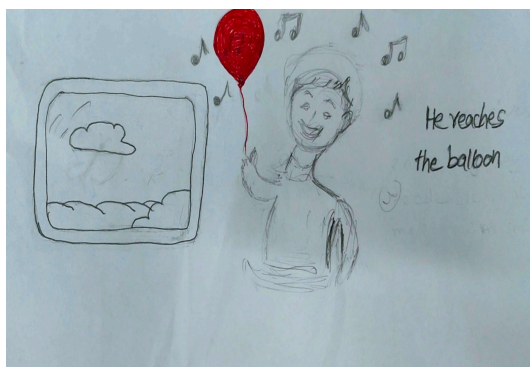
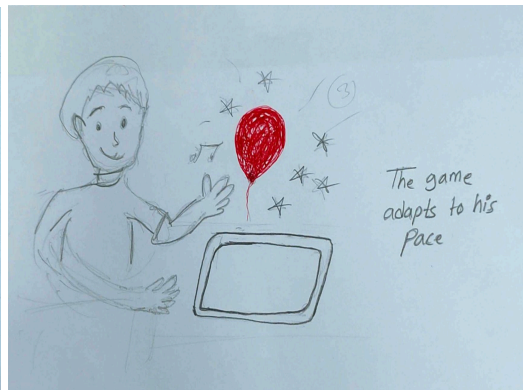
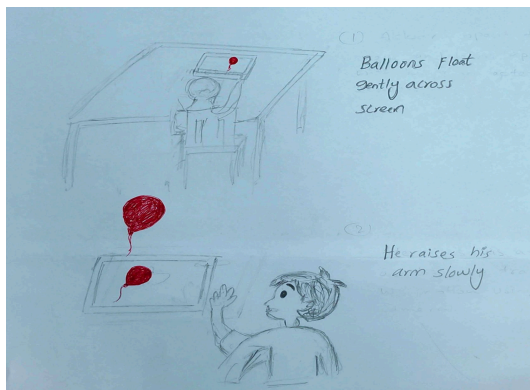
- Tracks patient progress over weeks
- Wants to assign tasks remotely
- Needs simple reports to share with parents and doctors

User Need Addressed:

Children with mild cerebral palsy who move their arms slowly need the game to adjust balloon movement speed to match their physical abilities.

Storyboard title: Catch the Balloon – Adaptive Speed**Explanation Paragraph:**

In this scene of the balloon-catching game, balloons float across the screen at speeds automatically adapted to the child's arm movement pace. For children who move their arms slowly, the balloons drift gently and slowly, providing ample time to reach out and catch them without feeling rushed or frustrated. This adaptive design empowers children with limited motor speed to participate successfully and enjoyably.



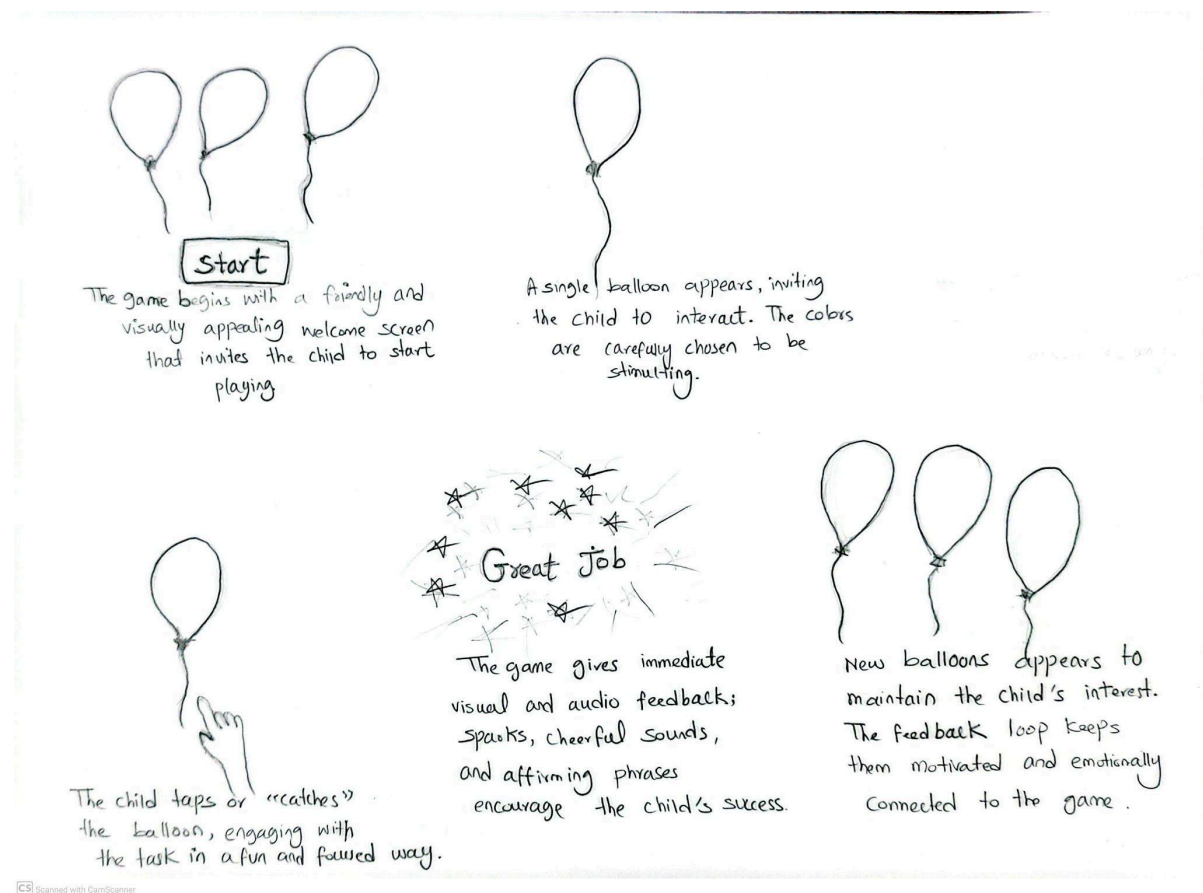
User Need Addressed:

Children with attention and motivation challenges respond better to colorful visuals and positive audio feedback that encourage ongoing participation.

Storyboard Title: Catch the Balloon – Sensory Feedback:

Explanation Paragraph:

When a child catches a balloon, the game immediately rewards the action with vibrant sparkles and joyful sound effects, along with encouraging phrases such as “Well done!” or “You’re amazing!” The balloons and background feature stimulating yet balanced colors to engage without overwhelming. This sensory feedback fosters motivation and emotional connection, helping children stay focused and enthusiastic.



User Need Addressed:

Children with limited attention spans benefit from short, structured activities with clear goals and opportunities to pause, reducing cognitive overload and physical fatigue.

Storyboard Title: Color Match Quest:**Explanation Paragraph:**

In this cognitive-focused balloon-catching scene, children are tasked with catching balloons of a specific color among multiple differently colored balloons floating on the screen. Each round introduces different target colors and varied numbers of balloons to catch, progressively increasing the challenge across levels. After completing a round, children can choose to take a break or continue, helping to reduce cognitive overload and physical fatigue. This structured and paced approach supports sustained focus, promotes color recognition skills, and keeps the game engaging and manageable.

